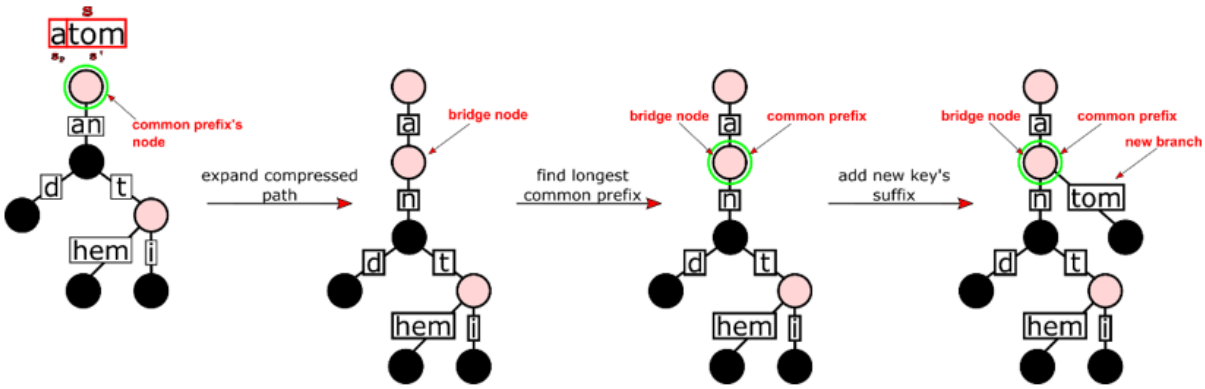


Figure 6.18 An example of method insert.

To better understand this operation, it might help you imagine that we decompress the edge to the child into a path, going back to the trie representation with one char per link. Then we traverse this path till we get to the end of the common prefix (to, say, a node B), we add a new branch as a child of B, and finally we compress again the two sub-paths on the two sides of B.

Listing 6.15 describes the pseudocode for the insert method, and figures 6.18 and 6.19 show two examples of how this method works on a simplified tree.

The method follows the same high-level logic as the trie's version we have seen in listing 6.5: we traverse the tree as far as we can (following the longest path covering a prefix of the string to insert), and the add a new branch for the new key.